

Grain Boundary Characterization of Carbide Ceramics for Fusion First-Wall Applications

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Nuclear fusion offers a promising pathway towards clean and high-density energy generation; however, the extreme conditions required to harness this energy hinder its practical feasibility. Ultra-high temperature ceramics (UHTCs) are therefore being studied for potential first-wall applications because of their stability under harsh conditions. Their performance, however, depends strongly on their microstructural properties, particularly grain boundaries. These interfaces between differently oriented crystals can weaken the material and serve as preferential sites where defects can accumulate. This work investigates the properties of grain boundaries and their relationships to material properties in four carbide-based ceramics: VC+SiC, VC+SiC+CF, WC+SiC+CF, and WC+ZrC+Ta+CF.

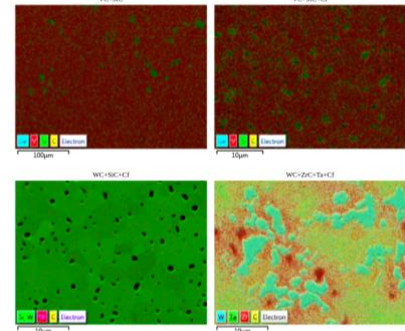


Figure 1 – Layered SEM-EDX elemental maps comparing four carbide composites.

Previous work by Lin et al. showed that the most compromising swelling of the irradiated ¹¹B-enriched ZrB₂ came from cracks along the grain boundaries¹. Samples with

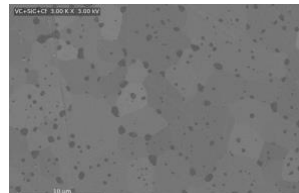


Figure 2 – SEM image of a VC composite with finely dispersed silicon carbide.

larger grains (approx. 18 μm) exhibited five times more grain-boundary cracking than samples with smaller grains (approx. 7 μm). In a related study, the same group found that adding fibers to similar ceramic materials could reduce cracking and swelling². These findings reveal grain boundaries as a key feature in understanding the strength of materials under extreme conditions.

Simulations and experiments were used to characterize the microstructure of carbide materials and examine features that may influence their response to extreme thermal and radiation conditions. Molecular simulations with RustBCA modeled radiation-induced atomic damage, and optical microscopy identified large-scale microstructural features and selected regions of interest for further analysis. These regions were imaged at higher resolution using scanning electron microscopy, and energy-dispersive X-ray spectroscopy mapped the spatial distribution of silicon carbide phases, including SiC particles located near or along grain boundaries (Figs. 1-2).

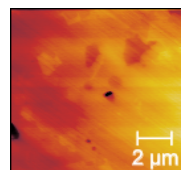


Figure 4 – AFM surface topography of WC.

Measurements of SiC particle size, spacing, grain size, and particle-boundary contact showed that finely dispersed second-phase SiC was associated with substantially smaller grains than agglomerated SiC, effectively restricting grain growth by producing a more stable microstructure with a greater density of interfaces (Fig. 3). Such interfaces can distribute deformation rather than allowing damage to concentrate within a small region of grains and boundaries. Atomic force microscopy was also used to obtain maps of surface friction and topography, corroborating the results from the SEM (Fig. 4). Together, the results reveal how finely dispersed SiC can stabilize the grain structure and increase the density of interfaces that may act as sinks for radiation-induced defects.

Finely dispersed second-phase SiC may therefore provide a practical means of controlling grain structure for fusion first-wall applications. Future work can involve irradiating the samples to directly test whether finely dispersed SiC suppresses grain coarsening and preserves microstructural stability for fusion first-wall applications.

	VC+SiC+CF dispersed	VC+SiC agglomerated
SiC second phase		
Mean particle radius, r (μm)	0.26	2.44
Mean spacing (μm)	1.3	10.9
Particles on grain boundaries (%)	29	90
Grains		
Measured grain size, d (μm)	10.5	82.6
Boundary density (μm^{-2})	0.80	0.06

Figure 3 – Comparison of vanadium carbide with dispersed & agglomerated SiC.

[1] Lin, Yan-Ru, et al. "Response of ¹¹B Enriched ZrB₂ Ultra-High Temperature Ceramic to Neutron Irradiation at Elevated Temperatures." *Acta Materialia*, vol. 276, 2024, article 120111, <https://doi.org/10.1016/j.actamat.2024.120111>.

[2] Lin, Yan-Ru, et al. "Comparative Neutron-Irradiation Effects on Thermal Conductivity Degradation and Dimensional Stability of TiC, TiB₂, and ZrB₂ at 200–1000 °C." *Journal of Nuclear Materials*, vol. 632, 2026, article 156897, <https://doi.org/10.1016/j.jnucmat.2026.156897>.